MIL-T-3901E
30 December 1981
SUPERSEDING
MIL-T-3901D
9 September 1977

MILITARY SPECIFICATION

TROUSERS, FIREMAN'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fireman's trouser with liner.

1.2 Classification. The trousers shall be of one type in the following sizes as specified (see 6.2):

Schedule of sizes

32 x 31	34 x 33	38 x 32	42 x 31
32 x 32	36 x 31	38 x 33	42 x 32
32 x 33	36 x 32	40 x 31	42 x 33
34 x 31	36 x 33	40 x 32	44 x 31
34 x 32	38 x 31	40 x 33	

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

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SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton
FF-R-556	-	Rivet, Solid, Small; Rivet, Split, Small; Rivet Tubular, Small; Burr; and Cap, Rivet; General Purpose
FF-W-92	-	Washers, Flat (Plain)
KK-L-271	-	Leather, Cattlehide, Strap, Vegetable Tanned
UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-286	-	Buttons, Tack; and Tack, Button
MIL-F-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-B-543	-	Buckles, Tongueless and Web Strap
MIL-T-3530	-	Thread and Twine; Mildew Resistant or Water Repellent Treated
MIL-S-6790	-	Suspenders, Trousers (Suspenders, Flying Suit)
MIL-F-10884	-	Fasteners, Snap
DOD-P-16232	-	Phosphate Coating, Heavy Maganese or Zinc Base (For Ferrous Metals)
MIL-C-43122	-	Cloth, Sateen, Cotton, Flame Resistant Treated
MIL-C-43410	-	Cloth, Coated: Polyester/Cotton, Polyester/Rayon; Polyvinyl Chloride or Polychloroprene Coated
MIL-C-43774	-	Cloth, Plain or Pajama Check Weave, Aramid
MIL-T-43636	-	Thread, Aramid
MIL-C-43708	-	Cloth, Coated or Bonded, Reflective
MIL-B-81813	-	Batting, Aramid or Nonvoloid Fiber, Quilted

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from the specification may appear in the sample, in which case, the specification shall govern.

* 3.2 Materials (see 6.4).

3.2.1 Basic material. The material for the outer shell shall be flame resistant treated cotton sateen cloth conforming to class 2 of MIL-C-43122.

* 3.2.2 Liner material. The material for the outer liner shall be a coated cloth conforming to MIL-C-43410. The color of the outer liner shall be natural or white.

* 3.2.3 Liner. The material for the liner shall be quilted batting conforming to MIL-B-81813. The color of the outer covering cloth shall be Olive Green 106 with the needle stitching thread of the dumbbell pattern (side facing wearer) being of an Olive Drab color and the bobbin thread (facing trouser) being a natural color thread. The minimum width of the batting shall be 24 inches.

* 3.2.4 Binding. The aramid cloth for binding waist, fly opening and leg bottoms shall be dyed OG-106, conforming to MIL-C-43774.

* 3.2.5 Reflectorized/fluorescent cloth. The reflectorized/fluorescent cloth shall conform to type I, class 3 or type II, class 2 of MIL-C-43708.

* 3.2.6 Thread. The thread for seaming and stitching, where required, shall be aramid, size E and B conforming to type I of MIL-T-43636.

* 3.2.6.1 Color.

* 3.2.6.1.1 Outer shell. The thread color for the outer shell shall be Olive Drab S-1 (C.A. 66022).

* 3.2.6.1.2 Liner. The thread color for the liner shall be Olive Drab S-1 (C.A. 66022), or be natural color.

* 3.2.6.1.3 Reflectorized/fluorescent strips. The thread color for the strips shall be natural.

* 3.2.6.2 Colorfastness. The Olive Drab S-1 (C.A. 66022) thread shall show fastness to laundering (after three cycles) equal to or better than the standard sample. When no standard sample is available, the colored thread shall show "good" fastness to laundering.

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* 3.2.7 Labels. Each trouser liner shall have a personal identification, size, identification and instruction label. All labels shall conform to DDD-L-20. The fastness to laundering requirements of DDD-L-20 shall apply.

* 3.2.7.1 Identification label. The identification label shall conform to type VI, class 1.

* 3.2.7.2 Size label. The size label shall conform to type VI, class 2.

3.2.7.3 Combination label. The identification label and the size label may be combined for the trouser liner.

* 3.2.7.4 Personal identification label. The personal identification label shall conform to type VI, class 10.

* 3.2.7.5 Instruction label. The instruction label shall conform to type VI, class 3, except that the contract number shall be included directly under the item description. The body of the instruction label shall be as follows:

TROUSERS, FIREMAN's

For maximum wearing comfort,
type 1 Suspenders, Trousers
(Suspenders, Flying Suit)
conforming to MIL-S-6790
(FSN 8440-00-275-5764) should
be worn with trousers.
Launder using Formula 1 of TM-10-280
DO NOT REMOVE THIS LABEL

Warning: In laundering, do not
use acid sour rinse. The rinse
bleaches out the flame proofing
agent in the garment

3.2.8 Leather. The leather for the straps and strap reinforcement pieces shall be vegetable-tanned cattlehide, type I, II, III or IV, class 2, 4/64 to 5/64 inch thick, conforming to KK-L-271. The color shall be either black or russet. The leather may be finished on the grain side with a light application of finish and dye, containing only sufficient pigment to assist in making the color uniform.

3.2.9 Snap fastener. The snap fastener for attachment of the outer shell to the trouser liner shall be a complete snap fastener (male and female), and shall consist of button, 24-line, size 1, and socket for the female half; and a size 1 eyelet with stud for the male half, conforming to style 2, finish 2 of MIL-F-10884.

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3.2.10 Rivets, tubular with caps or washers. The rivets shall be 1/8 inch nominal size, 3/16 and 3/8 inch minimal length, conforming to type XII, class 1, grade F of FF-R-556 except that they shall be drilled to the head and the inside of the rivet shank shall be chamfered. The caps shall conform to type II, grade F of FF-R-556. When washers are used in lieu of caps for securing the rivets, they shall fit the rivets and conform to type A, grade II, class E, of FF-W-92. The rivets, caps, and washers shall have a black chemical finish conforming to MIL-F-495. The 3/16 inch length rivet shall be used to attach the loop to the waistband and the clasp to the adjusting strap. The 3/8 inch length shall be used to attach the buckle strap and the adjusting strap to the waistband.

3.2.11 Buckle. The buckle shall be tongueless, two-bar, steel, conforming to type II, style 4, construction B, class 3, size 3/4 inch, of MIL-B-543.

3.2.12 Buttons. The buttons shall be type I, size 27-line with type IX tack, size 14-line by 0.406 inch prong length, conforming to MIL-B-286.

* 3.2.13 Clasp and take-up loop. The clasp and take-up loop shall be either brass conforming to figure 3, or steel conforming to figure 4 of this specification.

* 3.2.13.1 Finish.

* 3.2.13.1.1 Brass. The brass clasp and take-up loop shall have a black chemical finish conforming to MIL-F-495.

* 3.2.13.1.2 Steel. The steel clasp and take-up loop shall have a phosphate coating conforming to type M or Z, class 1 or 2 of DOD-P-16232.

* 3.3 Design. The trousers consist of an outer shell with a quilted aramid batting liner covered with a vapor barrier of rubber coated fabric. The outer trouser has a wide fly with snap fastener closure, waist adjustment strap with clasp and loop, two hip patch pockets with flaps and snap fasteners, and metal suspender buttons. The liner is attached permanently to the inside of the outer shell at the waistband and with snap fasteners at inseams and outseams of legs (figures 1 and 2). There are horizontal reflectorized/fluorescent cloth strips permanently stitched around the bottoms of the legs.

3.4 Figures. Figures 1 and 2 are furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figures, the written specification shall control.

* 3.5 Patterns. Standard patterns which provide an allowance of 3/8 inch for single needle and safety-stitch seams and 1/2 inch for double needle seams will be furnished by the Government. The Government patterns, which show directional lines and markings for proper assembly, shall not be altered in any way and are to be used for a guide for cutting the working patterns. The working patterns shall be identical to the Government patterns.

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3.5.1 List of pattern parts. The trouser shall be cut from the materials specified in accordance with the number of parts indicated in table I.

TABLE I. List of pattern parts

Material	Pattern nomenclature	Cut parts
	OUTER SHELL	
Flame resistant cotton sateen	Front	2
	Back	2
	Right fly	1
	Left fly	1
	Pocket	2
	Pocket flap	4
	Waistband lining	1
	LINING	
Coated cloth	Front	2
	Back	2
	LINER	
Aramid batting	Front	2
	Back	2
Leather	Buckle strap	1
	Adjusting strap	1

3.6 Construction.

* 3.6.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers or open seams will result. Where two or more methods for seams or stitch types are given for the same part of an operation, any one of them may be used. Where stitch type 401, 504 or 505 are used, the looper thread shall be on the inside of the trousers. All seams shall start and finish evenly. Unless otherwise indicated, the gage of stitching for the double-lapped, double-stitched seams shall be 3/16 to 5/16 inch. Unless otherwise indicated in table II, double-lapped, double-stitched seams may be lapped in either direction.

* 3.6.1.1 Type 301 stitching. Ends of all stitchings shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

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* 3.6.1.1.1 Repairs of type 301 stitching.

* 3.6.1.1.1.1 Quilted batting stitching. Broken stitches of the quilting pattern may be repaired by stitching back of each end of break not less than 1 inch.

* 3.6.1.1.1.2 Trouser and liner fabrication stitching.

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch in back of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

* 3.6.1.2 Type 401, 504 and 505 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. All repairs shall be in accordance with 3.6.1.1.1a. and 3.6.1.1.1b. Type 301 stitching may be used in repairing type 401 stitching.

* 3.6.1.3 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

* 3.6.2 Bartacking. Bartacking shall be 1/2 $\pm$ 1/16 inch in length and 1/8 + 1/32 inch in width and shall contain 28 stitches. Bartacking shall be free from thread breaks and loose or tight stitching.

* 3.6.3 Setting of snap fasteners. A hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than three splits shall occur in the button or eyelet barrels.

3.6.4 Repairs. Repairs such as mends, darns or patches shall not be made to the trouser assembly.

* 3.7 Manufacturing operations requirements. The trousers shall be manufactured in accordance with operation requirements specified in table II. The contractor is not required to follow the exact sequence of operations. Any additional basting or holding stitching used to facilitate manufacture is permissible, provided the thread is removed, or does not show on the finished garment.

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3.7.1 Shade and size marking. The cut parts of the trousers shall be marked to insure a uniform shade and size throughout the trouser. Any method of marking may be used except:

(a) Corrosive metal fastening devices.
(b) Sew-on shade tickets.
(c) Adhesive type tickets which leave traces of paper or adhesive on material after removal of the tickets.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting.						
	a. Cut trousers in accordance with pattern furnished which show directional lines, size, placement of pockets, and notches for proper assembly of all parts. Directional lines shall be placed in the warp direction.						
	b. The directional lines on the fronts and backs may vary from the warp direction by not more than 2-1/2 inches. Measurement shall be taken from the top and bottom of the directional line on pattern to selvage edge of the fabric and the difference between the two measurements shall not exceed 2-1/2 inches.						
	c. Cut all parts of the outer trouser out of the same piece of material except the underside of pocket flap, right fly, and waistband lining, which may be cut from ends.						
*	d. Cut quilted batting for the liner so that Olive Drab quilting needle thread faces the wearer and dumbbell pattern stitching runs lengthwise (warp direction).						
	NOTE: Care should be exercised when cutting the quilted batting and handling the cut parts in that the ends of the quilting stitching have a tendency to run back.						
	e. Cut the coated material for the liner so that in the finished liner the heavier coated side is on the outside.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd)						
	f. Cut leather reinforcement pieces for buckle strap and adjustment strap (one each) 1-1/2 to 1-3/4 inches long by 5/8 to 7/8 inch wide.						
*	g. Cut the aramid cloth for binding on the straight or bias or of sufficient width and lengths to meet the requirements specified in operation 22. Binding may be placed on rolls.						
	h. Cut a leather buckle strap and adjusting strap in accordance with patterns.						
2.	Replacement of damaged parts.						
	Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages, as classified in 4.2.2.1, are excluded and replaced with non-defective and properly matched material.						
3.	Shade marking.						
	a. Mark all parts, except those cut from ends, to insure a uniform shade and size throughout the outer trouser (see 3.7.1).						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
	Make outer trouser						
4.	Make two pocket flaps.						
	Stitch two pieces of flap together, trim corners, turn, work out points and edges and edge stitch with two rows of stitching, 1/4 to 5/16 inch apart, 1/16 to 1/8 inch from edge.	301 or 401 and 301	SSe-3(a) SSe-3(a) SSe-3(b)	8-10 8-10 8-10	E E E	E B E	
5.	Make two hip patch pockets and attach personal identification label.						
	a. Fold top of pocket as indicated by notches on pattern, and with raw edge turned in, single stitch 1/16 to 1/8 inch from edge, catching a stay of basic fabric of sufficient size to reinforce stud. The top edge of the personal identification label shall be caught in the hem of the right pocket. The stitching shall not be through the printing.	301	EFb-1	8-10	E	E	
	b. Single stitch the top folded edge of hem, 1/8 to 3/16 inch from edge. The finished hem shall measure 1 $\pm$ 1/4 inch.	301	OSf-1	8-10	E	E	
6.	Attach pockets.						
	Turn in raw edges of pockets and stitch to each back of trousers, at marks indicated on patterns, with two rows of stitching 1/4 to 5/16 inch apart, 1/16 to 1/8 inch from edge of pocket.	301	LSd-2	8-10	E	E	

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TABLE II. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
*7.	Attach flaps.						
	a. Stitch flaps to trousers with raw edge turned toward trouser to finish 1/2 ± 1/8 inch above pocket opening.	301	LSbk-3(a)	8-10	E	E	
*	b. Turn flap down over pocket and raise stitch with two rows of stitching 1/4 to 5/16 inch apart and 1/16 to 1/8 inch from folded edge.	301	LSbk-3(b)	8-10	E	E	
*8.	Make right fly lining.						
	a. Overedge the back edge of right fly lining.	502 or 503	EFd-1 EFd-1	8-10 8-10	B B	B B	
*	b. Fold fly lining (face to face) at notches indicated on pattern, stitch across bottom edge and turn.	301	SSa-1	8-10	E	E	
9.	Sew on right fly lining.						
	a. Seam right fly lining to right front from top to notch indicated on pattern and raise stitch, with seam allowance turned toward side of trousers.	301 or 401 and 301	LSq-2(a) LSq-2(a) LSq-2(b)	8-10 8-10 8-10	E E F	E B E	
	b. Fold fly lining as indicated by notch on pattern forming a 1-1/2 ± 1/8 inches extension at top tapering to 1 ± 1/4 inches at base of fly.						
	c. Single stitch the back edge of fly lining to the front of trousers from top to fly notch.	301	LSbj-1	8-10	E	E	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Make and attach left fly.						
	a. Fold front edge of left fly (face to face) at notches indicated on pattern and stitch across bottom from folded edge to within 1/4 inch of raw edge. Turn and force out corner.	301	SSa-1	8-10	E	E	
	b. Hem front edge with raw edge turned in. Finished hem shall measure 1-1/4 ± 1/8 inches.	301	EFb-1	8-10	E	E	
11.	Sew on left fly.						
	a. Join left fly to left front with a double-lapped, double-stitched seam.	301 or 401	LSc-2 LSc-2	8-10 8-10	E E	E B	
	- or -						
	b. Single stitch left fly to left front.	301 or 401	SSa-1 SSa-1	8-10 8-10	E E	E B	
	- and -						
*	c. Turn seam allowances on the underside toward side of trousers with raw edge turned in and stitch with two rows of stitching 1/4 to 5/16 inch gauge 1/16 to 1/8 inch from folded edge.	301		8-10	E	E	
12.	Join crotch seam.						
	a. Turn raw edge of crotch seam allowance of right front to the outside and single stitch 1/16 to 1/8 inch from folded edge.	301 or 401	LSdc-3(a) LSdc-3(a)	8-10 8-10	E E	E B	

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TABLE II. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Join crotch seam. (cont'd)						
*	b. Turn in raw edge of left front crotch seam allowance and stitch left front to right front with the two rows of stitching 1/4 to 5/16 inch apart, catching lower part of right extension in both rows of stitching.	301 or 401	LSdc-3(b) LSdc-3(b)	8-10 8-10	E E	E B	
*	c. Bartack end of crotch seam at base of fly with a 1/2 inch horizontal bartack.	Bartack		28 per bartack	B	B	
13.	Join seams.						
	a. Join seat seam with double-lapped, double-stitched seam.	401	LSc-2	8-10	E	B	
	b. Join outseams and inseams with double-lapped, double-stitched seams 1/4 to 5/16 inch gauge.	401	LSc-2	8-10	E	B	
14.	Sew on waistband lining.						
*	a. Stitch waistband lining to top of trousers with the rows of stitching, 1-1/2 ± 1/8 inches gauge, with the top row of stitching not more than 1/8 inch from top of trousers and with the top and bottom raw edges turned in. A two needle machine may be used. Finished waistband lining shall measure 1-3/4 ± 1/8 inches.	301 or 401	LSk-2 LSk-2	8-10 8-10	E E	E B	
*	b. Turn in each end of waistband lining and stitch to the edge of each fly.	301	SSb-1	8-10	E	E	

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					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Sew on waistband lining. (cont'd)						
	NOTE: The waistband lining may be attached before sewing down right fly (operation 9.c) and after attaching left fly but before hemming its front edge (see operation 10). When waistband lining is attached in this manner it shall be inserted far enough under the right fly lining and left fly hem to reinforce the snap fastener and take-up loop.						
15.	Hem bottoms.						
*	a. Turn up bottom of trousers, as indicated by marks on pattern and with raw edge turned in, single stitch 1/16 to 1/8 inch from edge. Finished hem shall measure 1 to 1-1/4 inches wide.	301	EFb-1	8-10	E	E	
*16.	Attach reflectorized material.						
	a. Position strip horizontally around bottom of leg 1-1/2 ± 1/4 inch from finished bottom to overlap 1/2 inch approximately at outseam. Stitch 1/16 to 1/8 inch from edges.	301 or 401	LSbj-1 LSbj-1	8-10 8-10	E E	E B	
17.	Attach fasteners.						
	a. Attach three stud (male) parts of fastener to the right fly and three socket (female) parts to the left fly as indicated by marks on pattern. The sockets must correspond to studs to effect a smooth closure.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Attach fasteners. (cont'd)						
*	b. Attach a socket part of fastener to the inseam and outseam of each leg, 4-1/4 ± 1/4 inches up from hemmed bottom, with the button part on outside of trouser.						
*	c. Attach a socket part of fastener to each flap 1/4 inch from point edges at point end. Attach a stud part to each pocket through the stay piece to correspond with stud and to provide a smooth closure.						
18.	Attach suspender buttons.						
	Attach eight suspender buttons on the outside of trouser vertically centered (1/4 inch off center tolerance) on waistband as follows:						
*	a. Two on each side of back, one 2-1/2 ± 1/4 inches from seat seam and one 1-1/2 ± 1/4 inches from first button toward outseam.						
*	b. One 1/4 inch toward front from each outseam (1/4 inch tolerance allowed toward front). Button should not be through double stitching.						
	c. One on each front 4-1/2 ± 1/4 inches from outseam.						
	NOTE: The dimensions shall be from center of seam to center of button.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Attach takeup loop, buckle strap, and adjusting strap						
*	a. Vertically position the takeup loop in the center of waistband on the inside left end and attach with two 3/16 inch rivets through waistband and waistband lining. An off center tolerance of 1/4 inch down from center of waistband will be allowed. Takeup loop shall not extend beyond end of left fly nor be more than 1/4 inch back of fly.						
*	b. Insert leather buckle strap over the back center bar of buckle, fold strap in half, ± 1/4 inch, and position on waistband of right back with the ends of strap of buckle even with marks on pattern (± 1/4 inch). For sizes 40 through 44 the rounded end of the buckle shall be aligned with the center of the outseam (± 1/4 inch). Attach to trouser with two 3/8 inch rivets through waistband, waistband lining, and leather reinforcement piece, positioned on inside of waistband. The rivets shall be uniformly centered with the first rivet 3/4 to 1 inch from folded end of strap and the second rivet 1 ± 1/8 inches from first rivet (measured from center to center).						
*	c. Centrally position clasp on adjusting strap (grain side of leather) with back edge of clasp 1/2 ± 1/8 inch in from straight edge of strap and attach with two 3/16 inch rivets.						

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
19.	Attach takeup loop, buckle strap, and adjusting strap. (cont'd)						
*	d. Vertically position adjusting strap to center of right waistband with clasp and takeup loop aligned and attach with two 3/8 inch rivets through adjusting strap, waistband, waistband lining and leather reinforcement piece positioned on inside of waistband. The rivets shall be uniformly centered with the first rivet 2-1/4 ± 1/2 inches from straight edge of adjusting strap and the second rivet 1-1/4 ± 1/2 inches from first rivet (measured from center to center). The clasp end of adjusting strap and the waistband at fly closure shall lie smooth and flat when the clasp is engaged.						
	Make trouser liner						
*20.	Overedge quilted batting edges.						
	Overedge fly openings, top, and bottom raw edges of fronts and backs.	504 or 505	EFd-1	6-12	B B	B B	
	NOTE: Face side of liner shall be identified by Olive Drab needle quilting thread and face wearer. Natural bobbin quilting thread shall face coated lining.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
21.	Join seams.						
*	a. Join front crotch seam from fly notch and back seat seam of the coated lining. Heavy coated side shall be face side (facing trousers).	401	SSa-1	8-10	E	B	
	b. Join outseams and inseams of the coated lining.	301 or 401	SSa-1 SSa-1	8-10 8-10	E E	E B	
*	c. Join front crotch seam from fly notch and seat seam of the quilted batting lining.	301 or 401	LSq-2(a) LSq-2(a)	8-10 8-10	E E	E B	
*	d. Turn crotch seam toward leg and seat seam toward other leg (to reduce bulk when stitching inseam seam), and stitch 1/4 to 5/16 inch from seam.	301 or 401	LSq-2(b) LSq-2(b)	8-10 8-10	E E	E B	
*	e. Join outseams and inseams of the quilted batting lining.	301 or 401	LSq-2(a) LSq-2(a)	8-10 8-10	E E	E B	
*	f. Turn seam allowances to one side and stitch 1/4 to 5/16 inch from seam. Crotch seams at inseams should not be staggered more than 1/4 inch.	301 or 401	LSq-2(b) LSq-2(b)	8-10 8-10	E E	E B	
22.	Bind coated fabric to quilted batting at waist, fly opening and bottoms.						
	NOTE: A holding stitch may be used to stitch coated material and batting edges together prior to binding. The stitching shall be 1/8 to 1/4 inch from edges and not show on finished liner						
	Finished appearance: The binding shall finish 3/8 to 5/8 inches wide.						

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TABLE II. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Bind coated fabric to quilted batting at waist, fly opening and bottoms. (cont'd)						
*	a. With coated material and overedged batting edges even (heavy coated side out), bind edges together stitching 1/16 to 1/8 inch from turned-in edges.	301	BSc-1	8-10	E	E	
	- or -						
	With face sides binding and batting together and edges even, stitch through binding, batting, and coated liner materials.	301 or 401	SSa-1 SSa-1	8-10 8-10	E E	E B	
*	b. Turn binding over edges, turn under raw edge, and stitch.	301	SSa-1	8-10	E	E	
*	c. The raw ends of binding shall be turned in and caught in stitching; or may be turned over binding to the outside of liner and tacked or bartacked. The fly binding shall overlap ends of waist binding.	301 or Bartack	SSa-1	8-10 28 per bartack	E B	E B	
*23.	Attach labels.						
	Position the personal identification and identfication label to left of the seat seam and the size label or combination label to the right of the seat seam on the batting side of liner 1 ± 1/4 inch from top of binding. Stitch around all edges. The stitching shall not be through the printing.	301 or 401	LSbj-1 LSbj-1	8-10 8-10	E E	E B	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
24.	Attach snap fasteners. Attach a stud part of snap fastener to each inseam and outseam of legs through coated fabric and quilted batting liner to correspond with socket parts on outer trouser (see 3.6.3).						
*25.	Attach liner to trouser shell. a. Insert liner into outer trouser. b. With fly binding folded over waist bindings coated liner side to inside of trouser, top edges of waists aligned, (liner shall not be visible from outside of trouser) right fly edge of liner 1/4 inch to inside of shell right fly, stitch 1/8 to 3/16 inch from top edge through all plies, backstitching end of stitching.	301	SSa-1	8-10	E	E	
	c. Bartack ends of stitching with a vertical bartack through all layers catching ends of stitching.	Bartack		28 per bartack	E	E	
	d. The waist and legs on liner shall fit smoothly inside the outer trouser.						
26.	Clean trousers (outer trouser and liner). a. Trim all thread ends and remove loose threads. b. Remove all spots, stains and shade markings.						

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* 3.8 Measurements. The finished trousers shall conform to measurements shown in table III.

TABLE III. Table of measurements in inches

	Outer trousers		Trouser liner	
Size	1/2 Waist 1/	Inseam 2/	1/2 Waist 3/	Inseam 4/
32 x 31	17	31	16-3/4	27
32 x 32	17	32	16-3/4	28
32 x 33	17	33	16-3/4	29
34 x 31	18	31	17-3/4	27
34 x 32	18	32	17-3/4	28
34 x 33	18	33	17-3/4	29
36 x 31	19	31	18-3/4	27
36 x 32	19	32	18-3/4	28
36 x 33	19	33	18-3/4	29
38 x 31	20	31	19-3/4	27
38 x 32	20	32	19-3/4	28
38 x 33	20	33	19-3/4	29
40 x 31	21	31	20-3/4	27
40 x 32	21	32	20-3/4	28
40 x 33	21	33	20-3/4	29
42 x 31	22	31	21-3/4	27
42 x 32	22	32	21-3/4	28
42 x 33	22	33	21-3/4	29
44 x 31	23	31	22-3/4	27
Tolerance	± 1/2	+ 3/4 - 1/2	± 1/2	+ 3/4 - 1/2

1/ 1/2 waist. With fly closed and trousers smooth and flat measure through center of waistband from folded edge to folded edge.

2/ Inseam. Measure inseam from crotch seam to bottom of leg.

3/ 1/2 waist. With liner pulled out of trousers, and front fly edges even along binding of waist from folded edge to folded edge.

3.9 Workmanship. The finished trousers shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When a certificate of compliance is submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in the specification or applicable purchase document. The contractor shall furnish a certificate of compliance stating that the clasp and take-up loop conform to the requirements of 3.2.13.

* 4.2.2 End item examination. The end item shall be examined in accordance with 4.2.2.1 and 4.2.2.2. The lot size shall be expressed in units of one trouser with liner. The sample unit shall be one trouser with liner.

* 4.2.2.1 Visual examination. The end item shall be examined for the defects listed below. The inspection level shall be II and the acceptable quality level (AQL) shall be 4.0 major, 15.0 major and minor A combined, and 40.0 total (major, minor A, and minor B combined) defects per hundred units.

Defects	Classification Major	Minor A	Minor B
Material defects and damages			
a. Hole, cut, tear or needle chew:			
1. In outershell: - on outside	X		

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Defects	Classification Major	Minor A	Minor B
Material defects and damages (cont'd)			
a. Hole, cut, tear or needle chew: (cont'd)			
2. In liner:			
- any on outside layer of liner or through both layers of liner	X		
- any on inside layer of liner		X	
b. Visible mend, darn, or patch:			
1. On outershell:			
- any		X	
2. On liner:			
- up to 1/4 inch (inclusive) in diameter or length			X
- more than 1/4 inch in diameter or length		X	
c. Weakening defects:			
1. Smash	X		
2. Multiple float or loose slub:			
- on outershell		X	
- on liner			X
d. Quilted batting:			
1. Broken or missing stitching not repaired:			
- 1 inch or more	X		
- less than 1 inch			X
2. Batting filler showing face side of liner (After assembly to coat, facing wearer)			X
e. Permanent crease in cloth (when caught in seaming)		X	
f. Unsightly slub or shade bar:			
- on outershell		X	
- on liner			X
g. Any outside part of outershell shaded			X

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Defects	Classification Major	Minor A	Minor B
Cleanness			
a. Any spot or stain: - on outside of outershell or liner			X
b. One or more shade tickets not removed			X
c. Any shade marking visible on outside of outershell or liner		X	
d. Thread ends not trimmed or loose threads not removed throughout			X
Component and assembly			
Any component part or required operation omitted (unless otherwise classified herein)	X		
Trouser liner not lined with uncoated side of coated material facing liner	X		
Trouser liner coated material facing wearer after assembly	X		
Cutting			
a. Any component part not cut in conformance with directional lines on patterns or in accordance with specification requirement		X	
b. Directional line on front or back varying from warp direction by more than 2-1/2 inches	X		
Seams and stitching			
a. Accuracy of seaming:			
1. Seam(s) twisted, puckered, or pleated (unless otherwise classified herein)			X

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Defects	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
2. Part of material caught in unrelated operation or stitching:			
- up to a distance of 1/4 inch inclusive		X	
- for a distance greater than 1/4 inch	X		
3. Ends of stitching when not caught in other seams or stitching backtacked less than 1/4 inch		X	
4. Stitching overlapping the end(s) of thread break less than 1/2 inch			X
5. Ends of a continuous line of stitching overlapped less than three stitches on labels or less than 1/2 inch on all other stitching			X
6. Ends of stitching for reflectorized material overlapped less than 1 inch			X
7. Various shades of thread used on outside of same liner		X	
b. Gage of stitching and seam allowance width:			
1. Irregular, i.e., unevenly gaged or corresponding stitchings not uniform (to be scored only when this condition exists on major portion of seam)		X	
2. Not within range of gage specified, or varies more than 1/16 inch when no range is specified			X
c. Stitches broken or missing:			
1. All stitching (except overedging):			
- up to 1/2 inch inclusive			X
- more than 1/2 inch	X		
2. Overedging:			
- over 1/2 inch			X
d. Skipped stitches:			
- 1/4 to 1/2 inch long			X
- more than 1/2 inch long		X	

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Defects	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)			
e. Runoff:			
- 1/4 to 1/2 inch long			X
- more than 1/2 inch long		X	
f. Raw edge:			
1. On outside:			
- more than 1/4 inch up to 1/2 inch inclusive			X
- more than 1/2 inch		X	
2. On inside:			
- 1/2 to 1 inch inclusive			X
- more than 1 inch		X	
g. Seam and stitch type:			
1. Seam type or stitch type not as specified		X	
2. Looper thread on 401 stitch type on outside of garment		X	
h. Stitches per inch (to be scored only when condition exists on major portion of seam of stitching):			
- more than maximum specified in shell or liner		X	
- less than minimum specified		X	
- more than specified to attach reflectorized strips	X		
i. Stitch tension (except bartack):			
1. Loose tension resulting in a loose seam	X		
2. Loose tension (exposed loops of top or bottom thread):			
- for a distance of 1/2 inch or more, on outside			X
3. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching):			
- seat, outseams, inseams and crotch seam:			
(a) Both rows of stitching at same place	X		
(b) Both rows of stitching at different place		X	
(c) One row of stitching only			X

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Defects	Classification Major	Minor A	Minor B
Seams and stitches (cont'd)			
NOTE: Puckering is evidence of tight tension. When puckering is evident the seam shall be tested by exerting normal pull in the lengthwise direction of seam. Aramid quilted fabric has a tendency to pucker at joining seams and at stitchings. The puckering when not so excessive as to seriously affect appearance shall not be classified as a defect.			
j. Bartacks			
1. Missing, misplaced or insecure, failing to serve intended purpose		X	
2. Loose tension		X	
3. Less than 7/16 or more than 9/16 inch in length			X
Metal components:			
1. Broken or malformed affecting functions	X		
2. Burrs or sharp edges		X	
3. Improperly clinched failing to effect a secure closure		X	
Fly			
a. Snap fastener out of alignment causing bulge or twist		X	
b. Hem of left fly less than 1-1/8 or more than 1-3/8 inches wide			X
c. Right fly extension:			
- less than 1-3/8 or more than 1-5/8 inches at top			X
- less than 3/4 or more than 1-1/4 inches at base			X

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Defects	Classification Major	Minor A	Minor B
Waistband			
a. Width of waistband lining less than 1-5/8 or more than 1-7/8 inches			X
b. Suspender button(s) not positioned as specified		X	
c. One or more suspender buttons omitted	X		
d. Takeup loop or buckle strap not positioned as specified		X	
e. Takeup loop affixed to outside of waistband			X
f. Buckle strap not folded as specified			X
g. Buckle strap not positioned as specified			X
h. Leather reinforcement(s) omitted		X	
i. Buckle strap or adjustment strap attached with less than or more than two rivets		X	
j. Clasp and takeup loop not aligned			X
k. Clasp end of adjustment strap does not lie flat when fly is closed and clasp and loop are engaged		X	
Labels			
a. Missing, incorrect or illegible: - instruction label		X	
- size label	X		
- personal identification label		X	
b. Not positioned as specified			X
c. Instruction label or size label not stitched on all four sides		X	

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Defects	Classification Major	Minor A	Minor B
Pockets and flaps			
a. Width of pocket hem less than 3/4 inch or more than 1-1/4 inches			X
b. One or more snap fasteners or snap fastener part omitted		X	
c. Stud part of snap fastener not attached through stay piece reinforcement		X	
d. Socket part of fastener in flap not corresponding to stud part in pocket resulting in pucker or twist		X	
Hems			
Width of hem less than 1 inch or more than 1-1/4 inches			X
Coated fabric			
a. Area of no coating, blister or peel: - up to 1/4 inch in diameter inclusive - over 1/4 inch in diameter	X	X	
b. More than one coating repair on a liner		X	
c. Tacky coating	X		
d. Lighter coated side of fabric on the outside of liner		X	
Front opening			
a. Snap fastener in reverse position (stud part on left side and socket part on right side)			X
b. Overlap at top less than 3/4 or more than 1-1/4 inches			X
Seams (liner)			
a. Bottoms, waist and fly edges of component:			
1. Overedge stitching omitted on two or more seams		X	
2. Overedge stitching omitted on edges of liner or bottom of sleeves prior to binding operation		X	

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Defects	Classification Major	Minor A	Minor B
Seams (liner) (cont'd)			
3. Overedge stitching on one seam omitted		X	
4. Overedge stitching not caught through all plies of material for a distance of 1 inch or more			X
b. Binding:			
1. Not securely caught on fabrics causing binding to pull away from fabric		X	
2. Twisted, distorted or puckered, binding stretched, i.e., edges have a wavy appearance			X
3. Tight, causing edges of liner to curl			X
4. Ends not turned under and caught in the stitching or turned to the outside and tacked		X	
5. Finishes less than 3/8 or more than 5/8 inch wide			X
Assembly			
a. Snap fasteners:			
- Not properly aligned at legs, causing distortion when trouser and liner are fastened together	X		
- Any fastener not functioning properly i.e., fails to snap closed, provide a secure closure or to open freely	X		
NOTE: The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.			
- Clinched excessively tight, cutting adjacent material	X		
- Clinched loosely, permitting any component to rotate freely but not to the degree that any component can be expected to become detached during use			X
- Clinched loosely to the degree that components can be expected to become detached during use		X	

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Defects	Classification Major	Minor A	Minor B
Assembly (cont'd)			
NOTE: Incomplete roll of end of button eyelet barrel is evidence of improper and insecure clinching.			
- Incorrect style		X	
- More than three splits in eyelet or button barrels			X
b. Trouser and liner not the same size	X		
c. Trouser and liner incorrectly attached at waist causing distortion		X	

* 4.2.2.2 Dimensional examination. The end item shall be examined for the defects listed below. Any measurement deviating from any dimension and tolerance specified in table III shall be scored as a size measurement defect. The inspection level shall be S-3 and the AQL shall be 4.0 defects per hundred units.

4.2.3 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flap, improper taping, loose strapping, inadequate stapling. Bulged or distorted container.
Content	Number of bundles per shipping container is more or less than required. Number of trousers per bundle is more or less than required. 1/ Size shown on one or more trousers not as specified on container. 1/

1/ For this defect one bundle from each container in the sample shall be examined.

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5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Level A. Each pair of trousers with attached quilted liner shall have one side superimposed on the other with inseams meeting and be folded from the leg bottom up toward the waistband to an overall length of approximately 23 inches. For the larger sizes, it may be necessary to adjust the seat area to achieve the desired width of approximately 14-1/2 inches. Five pairs of trousers of one size only shall be neatly stacked in a bundle with the waistbands alternated and securely tied at each end with cotton tape or twine.

* 5.1.2 Commercial. Trousers shall be preserved in accordance with MIL-STD-1188.

* 5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Ten pairs of trousers of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container, conforming to style RSC-L, grade V2s of PPP-B-636. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A bundles shall be packed flat two in depth within a shipping container. Inside dimensions of each shipping container shall be 23-1/2 inches in length, 15 inches in width, and 15 inches in depth (± 1/8 inches). Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.2.3. Toward the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

* 5.2.2 Level B packing. Ten pairs of trousers of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container,conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A bundles shall be packed flat two in depth within a shipping container. Inside dimensions of each shipping container shall be 23-1/2 inches in length, 15 inches in width, and 15 inches in depth (± 1/8 inches). Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.2.3. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

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* 5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.3.

* 5.2.3 Commercial packing. Trousers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

* 5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantities and NSN's contained therein.

6. NOTES

6.1 Intended use. The trousers are intended for wear by firefighting personnel of the Department of Defense.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Sizes required (see 1.2).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to sample of end item and shade samples, address the procuring activity issuing the invitation for bids.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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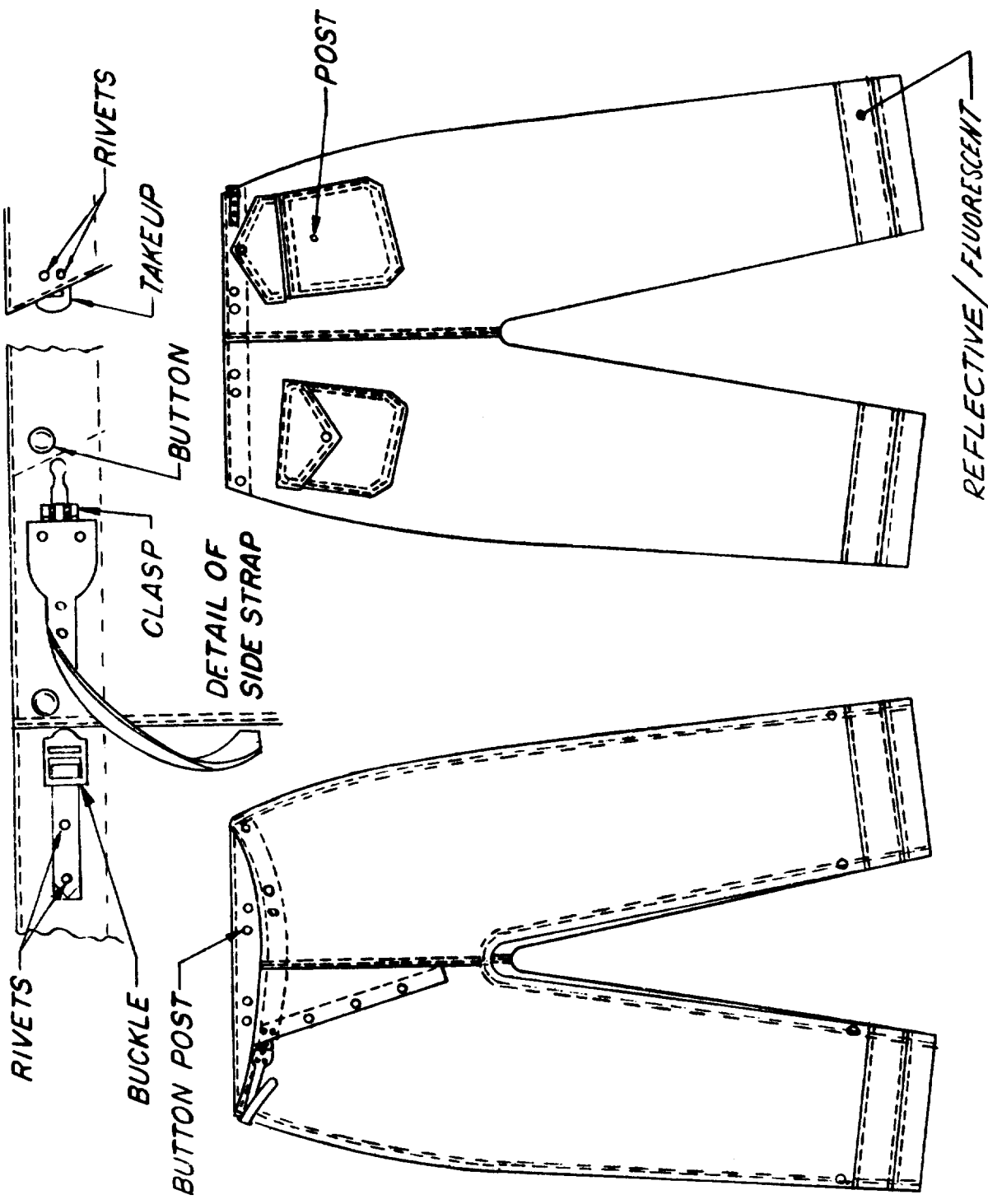


FIG. 1 – TROUSERS, FIREMAN'S, (OUTER TROUSER)

2-1-290

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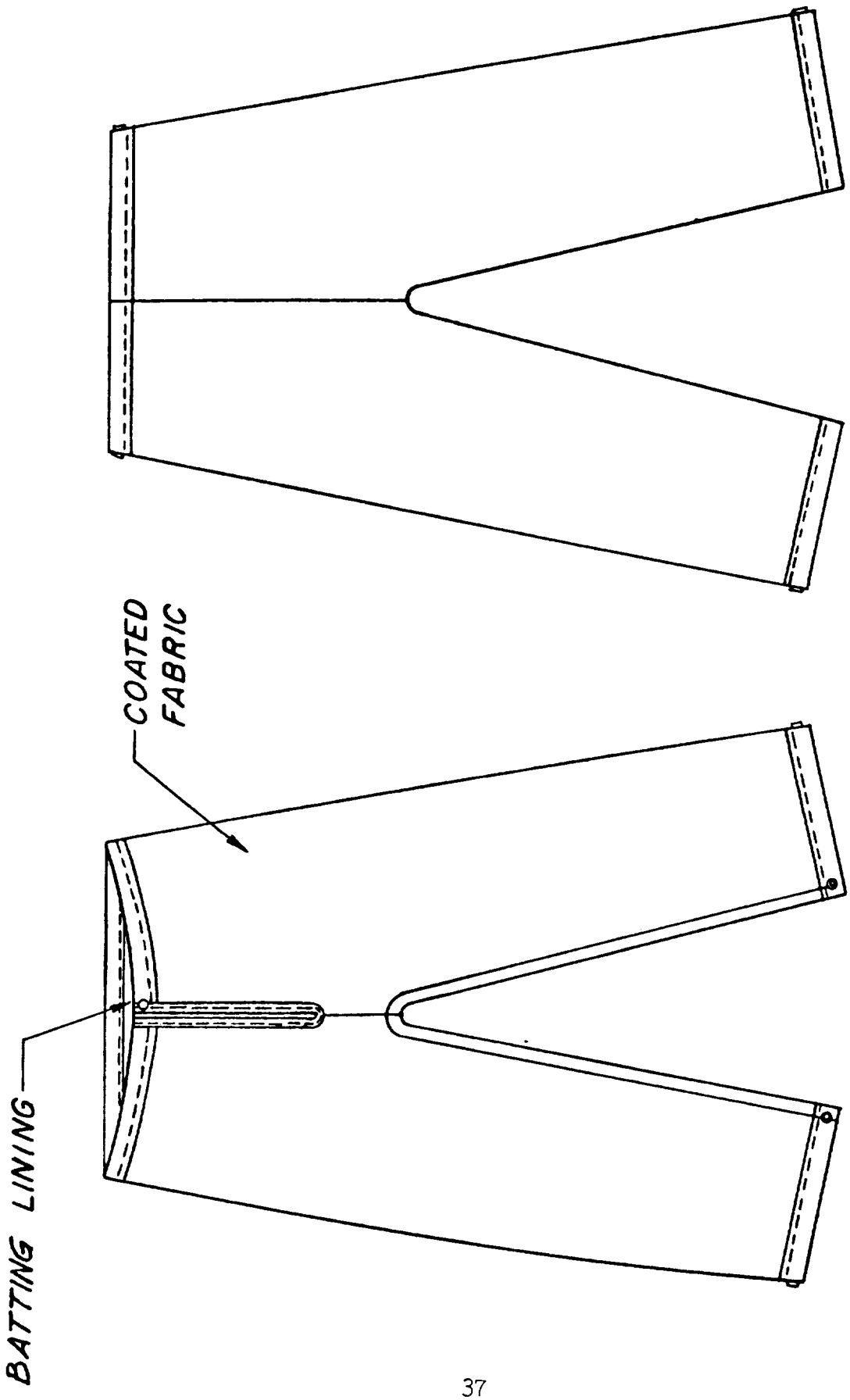
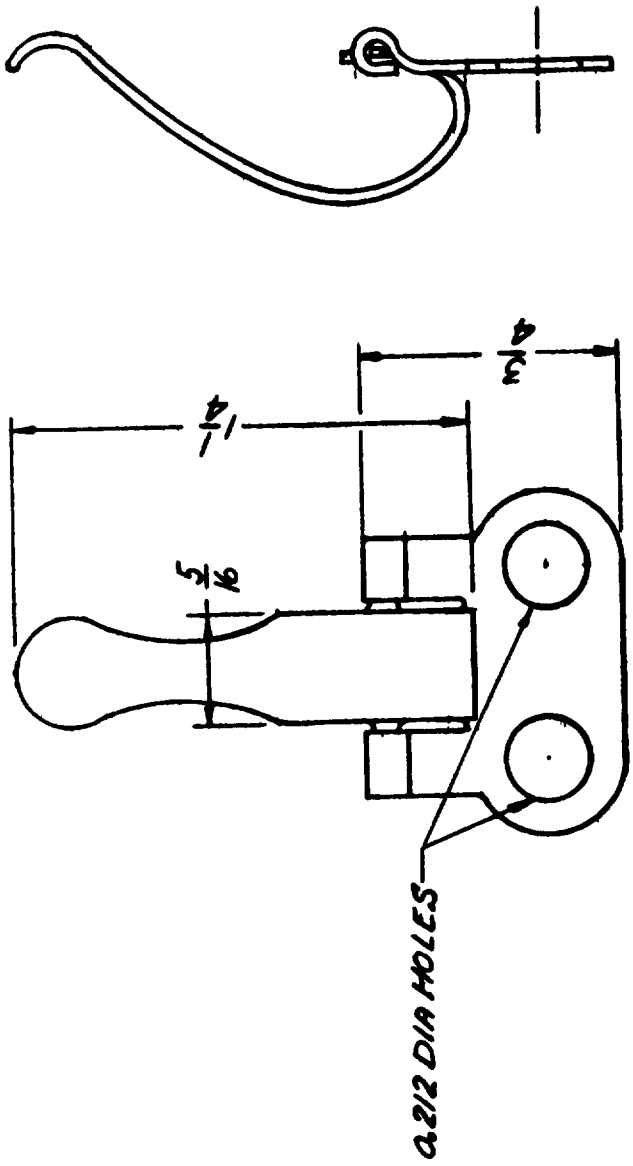


FIG. 2 - TROUSERS, FIREMAN'S, (LINER) 2-1-295

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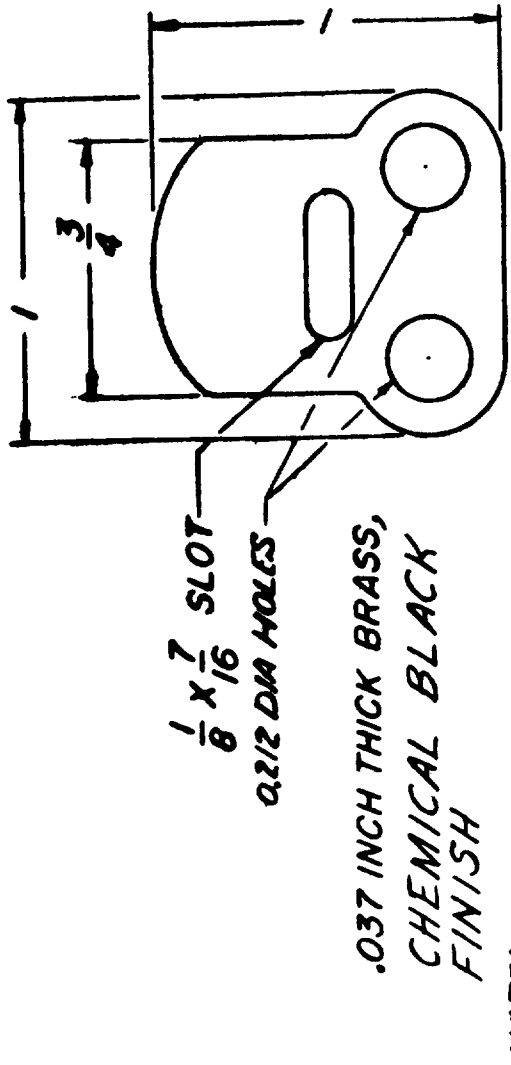
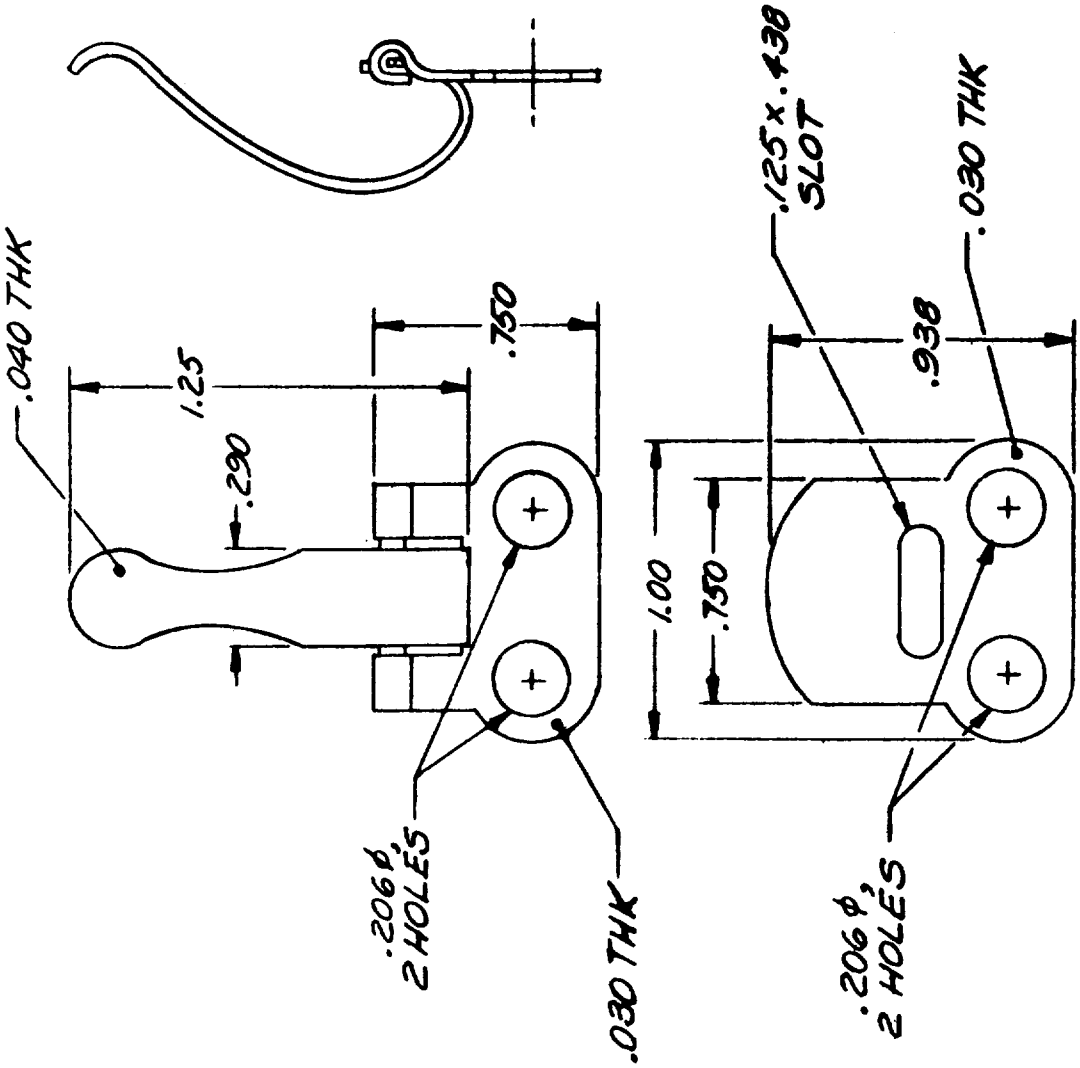


FIG 3 - CLASP & TAKE-UP LOOP (BRASS)
4-1-148

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NOTES:
1. ALL DIMENSIONS IN INCHES
2. MATL: STEEL, PHOSPHATE COATED

FIG. 4 CLASP & TAKE-UP LOOP (STEEL)
2-1-1577

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